Blockchain : Experiment – 6

Aim:

Creation and Deployment of Smart Contract and Transactions using Solidity and Remix IDE (Voting Ballot).

Theory:

Relevance of Required Statements in Solidity Programs:

In Solidity, the required statement acts as a guard condition within functions. It ensures that only valid inputs or authorized users can execute certain parts of the code. If the condition inside require is not satisfied, the function execution stops immediately, and all state changes made during that transaction are reverted to their original state. This rollback mechanism ensures that invalid transactions do not corrupt the blockchain data.

For example, in a Voting Smart Contract, require can be used to check:

- Whether the person calling the function has the right to vote (require(voters[msg.sender].weight > 0, "Has no right to vote");).
- Whether a voter has already voted before allowing them to vote again.
- Whether the function caller is the chairperson before granting voting rights.

Thus, require statements enforce security, correctness, and reliability in smart contracts. They also allow developers to attach error messages, making debugging and contract interaction easier for users.

Keywords: mapping, storage, and memory

1. mapping:
   A mapping is a special data structure in Solidity that links keys to values, similar to a hash table. Its syntax is mapping(keyType => valueType). For example:
   mapping(address => Voter) public voters;
   Here, each address (Ethereum account) is mapped to a Voter structure. Mappings are very useful for contracts like Ballot, where you need to associate voters with their data (whether they voted, which proposal they chose, etc.). Unlike arrays, mappings do not have a length property and cannot be iterated over directly, making them gas efficient for lookups but limited for enumeration.
2. storage:
   In Solidity, storage refers to the permanent memory of the contract, stored on the Ethereum blockchain. Variables declared at the contract level are stored in storage by default. Data stored in storage is persistent across transactions, which means once written, it remains available unless explicitly modified. However, because writing to blockchain storage consumes gas, it is more expensive. For example, a voter's information saved in the voters mapping remains available throughout the contract's lifecycle.
3. memory:
   In contrast, memory is temporary storage, used only for the lifetime of a function call. When the function execution ends, the data stored in memory is discarded. Memory is mainly used for temporary variables, function arguments, or computations that don't need to be permanently stored on the blockchain. It is cheaper than storage in terms of gas cost. For instance, when handling proposal names or temporary string manipulations, memory is often used.

Thus, a smart contract developer must balance between storage and memory to ensure efficiency and cost-effectiveness.

Why bytes32 instead of string?
In earlier implementations of the Ballot contract, bytes32 was used for proposal names instead of string. The reason lies in efficiency and gas optimization.

- bytes32 is a fixed-size type, meaning it always stores exactly 32 bytes of data. This makes storage simple, comparison operations faster, and gas costs lower. However, it limits proposal names to 32 characters, which is not very flexible for user-friendly names.
- string is a dynamically sized type, meaning it can store text of variable length. While it is easier for developers and users (since names can be written normally), it requires more complex handling inside the Ethereum Virtual Machine (EVM). This increases gas usage and may slow down comparisons or manipulations.

To make the system more user-friendly, modern implementations of the Ballot contract often convert from bytes32 to string. Tools like the Web3 Type Converter help developers easily switch between these two types for deployment and testing.

In summary, bytes32 is used when performance and gas efficiency are priorities, while string is preferred for readability and ease of use.

Code :

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract VotingBallot {

    struct Candidate {
        string name;
        uint256 voteCount;
    }

    Candidate[] public candidates;

    mapping(address => bool) public hasVoted;

    address public owner;

    event CandidateAdded(
        uint256 indexed candidateId,
        string name
    );

    event VoteCast(
        address indexed voter,
        uint256 indexed candidateId
    );

    modifier onlyOwner() {
        require(
            msg.sender == owner,
            "Only owner can perform this action"
        );
        _;
    }

    constructor() {
        owner = msg.sender;
    }

    function addCandidate(
        string memory _name
    ) public onlyOwner {

        require(
            bytes(_name).length > 0,
            "Candidate name cannot be empty"
        );

        candidates.push(
            Candidate({
                name: _name,
                voteCount: 0
            })
        );

        emit CandidateAdded(
            candidates.length - 1,
```

```
        _name
    );
}

function vote(
    uint256 _candidateId
) public {

    require(
        !hasVoted[msg.sender],
        "You have already voted"
    );

    require(
        _candidateId < candidates.length,
        "Invalid candidate"
    );

    hasVoted[msg.sender] = true;

    candidates[_candidateId].voteCount++;

    emit VoteCast(
        msg.sender,
        _candidateId
    );
}

function getCandidateCount()
    public
    view
    returns (uint256)
{
    return candidates.length;
}

function getCandidate(
    uint256 _candidateId
)
    public
    view
    returns (
        string memory,
        uint256
    )
{
    require(
        _candidateId < candidates.length,
        "Invalid candidate"
    );

    Candidate memory candidate =
        candidates[_candidateId];

    return (
        candidate.name,
        candidate.voteCount
    );
}

function getWinner()
    public
    view
    returns (
        string memory winnerName,
        uint256 winnerVotes
    )
{
    require(
        candidates.length > 0,
        "No candidates available"
    );

    uint256 winningIndex = 0;

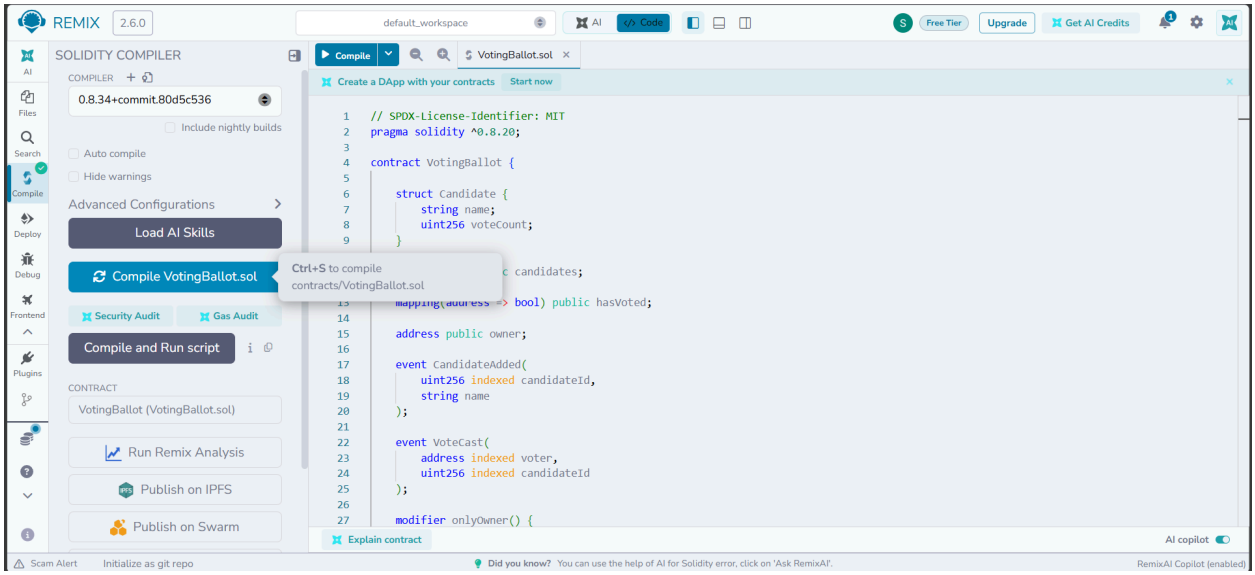
    for (
        uint256 i = 1;
        i < candidates.length;
        i++
    ) {
        if (
            candidates[i].voteCount >
            candidates[winningIndex].voteCount
        ) {
            winningIndex = i;
        }
    }

    winnerName =
        candidates[winningIndex].name;

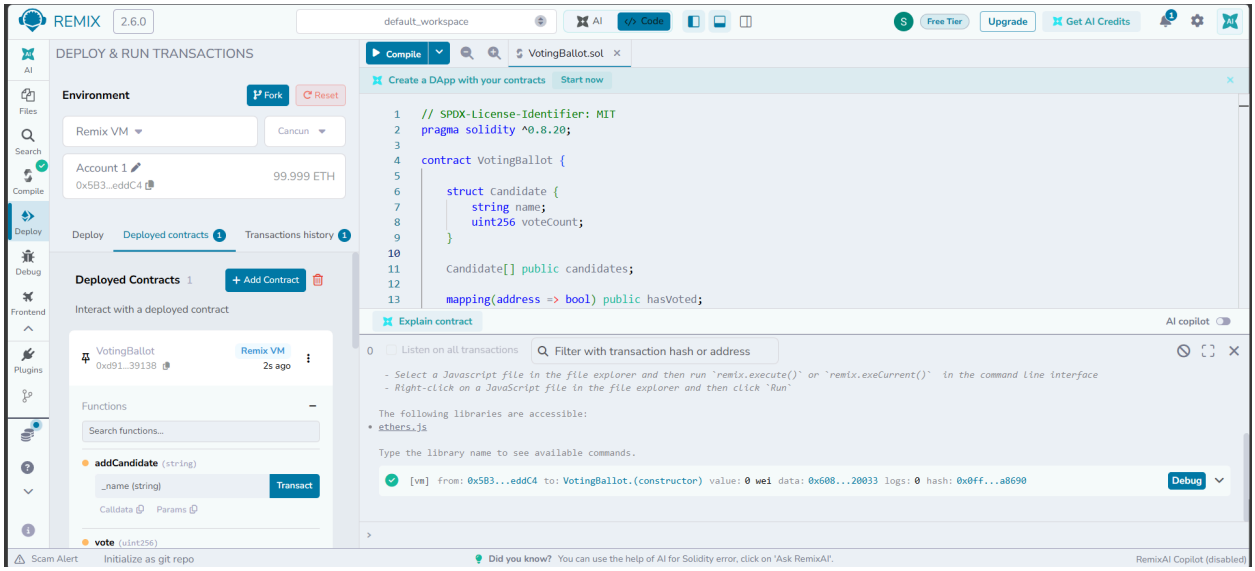
    winnerVotes =
        candidates[winningIndex].voteCount;
}
}
```

Output:

Contract Compiled:



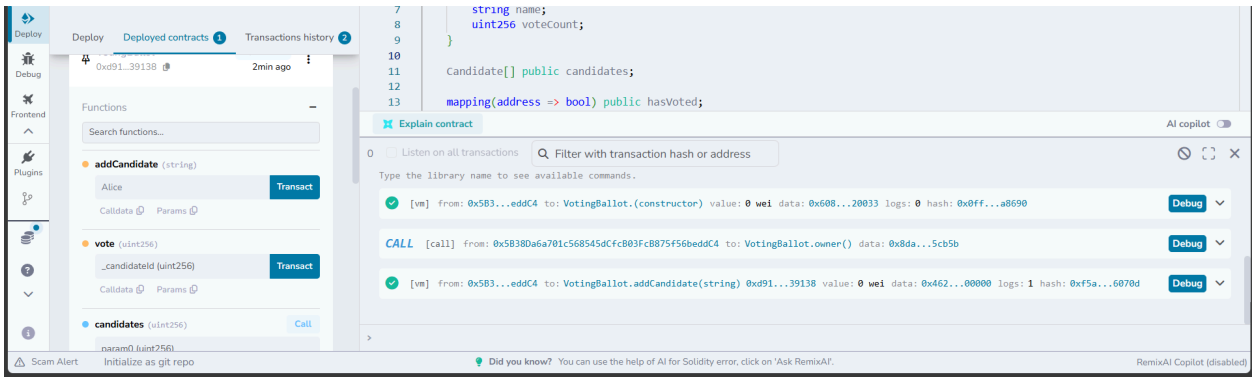
Contract Deployed:



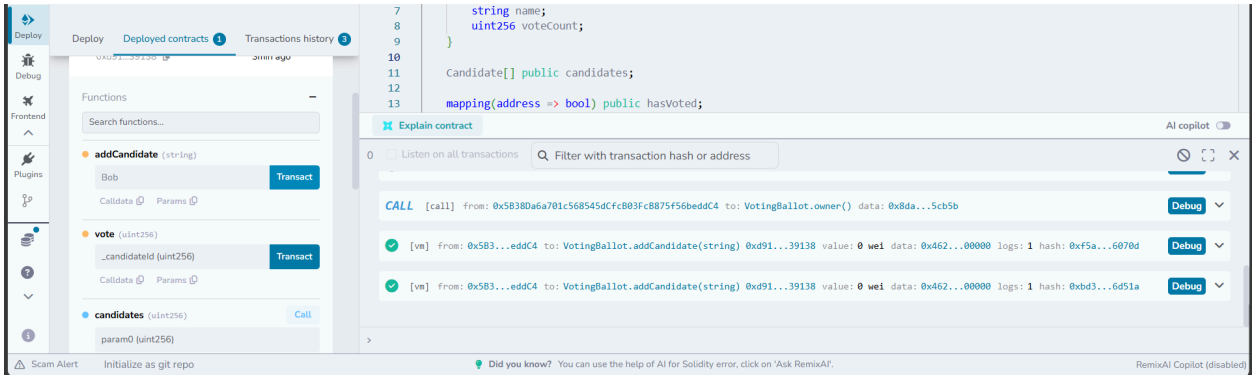
Owner of contract:



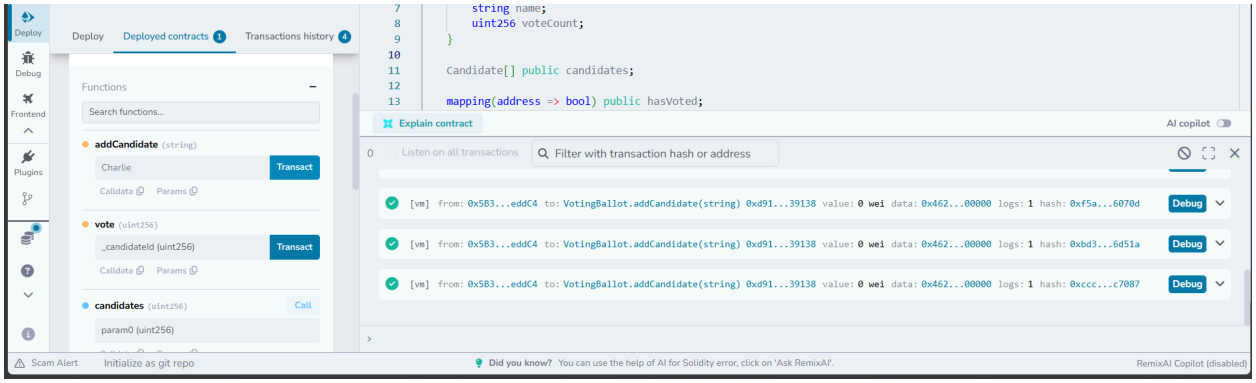
Added Alice as candidate:



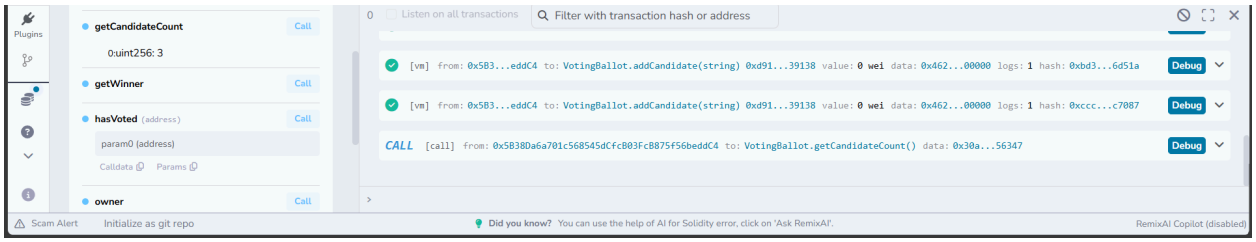
Added 2nd Candidate Bob:



Added 3rd candidate Charlie:



Calling getCandidateCount:



The output is right as its 3.

Calling getCandiate function on Alice:



Alice has 0 votes.

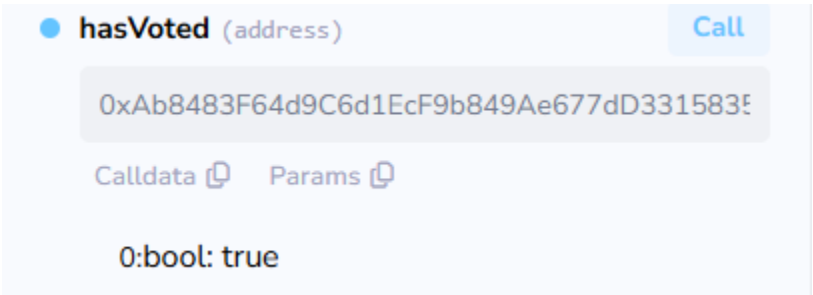
Voting for Alice using Account 2:



Verify that the vote is given to Alice:



Verify Account 2 has voted:



Preventing double voting:



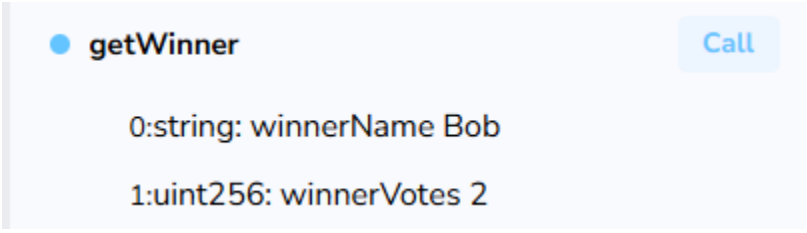
Double voting prevented successfully

Using 3rd and 4th account to vote for Bob(1):





Finding the winner using getWinner:



Conclusion:

In this experiment, a Voting/Ballot smart contract was deployed using Solidity on the Remix IDE. The concepts of require statements, mapping, and data location specifiers like storage and memory were explored to understand their role in ensuring security, efficiency, and correctness in smart contracts. The difference between using bytes32 and string for proposal names was also studied, highlighting the trade-off between gas efficiency and readability. Overall, the experiment provided practical insights into the design and deployment of voting contracts on the blockchain